

Zihan Liang

Undergraduate AI Researcher at Xi'an Jiaotong-Liverpool University

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Research Profile

Undergraduate researcher in artificial intelligence developing reproducible machine-learning methods for mixed-frequency time-series forecasting and sequential social-media risk assessment. Experience spans Transformer and attention architectures, NLP sequence modeling, multi-horizon evaluation, imbalanced classification, and interpretability. Research communication includes an SSRN preprint and an ICSC conference poster.

Education

Xi'an Jiaotong-Liverpool University (XJTLU)	Sep 2024 – Jul 2028, expected
BEng Artificial Intelligence (Intelligent Systems)	Suzhou, China
Dual Degree: BSc (Hons), University of Liverpool, expected	Liverpool, UK
– Academic standing: Completed Stage 2 with a Stage 2 weighted average of 78/100 and an overall weighted average of 74/100.	
– Selected marks: Python for AI 92, Maths for Machine Learning 86, Engineering Mathematics I 82, AI Fundamentals & Ethics 81, and Data Structures and Algorithms 78.	

Research Outputs

- Q. Luo, R. Xu, **Z. Liang**, Z. Hou, and R. Zhao. *Do Macroeconomic and Geopolitical Risk Factors Improve FX Volatility Forecasts? A Mixed-Frequency Transformer Approach*. SSRN preprint, 2025. [Preprint](#) | [DOI](#).
- **Z. Liang**, R. Xu, and R. Zhao. *Public Mental Health Monitoring: Detecting Suicide Risk in Sequential Social Media*. Conference poster presented at ICSC 2025. [Code](#) | [Poster](#).

Research Experience

Mixed-Frequency Transformer for USD/CNH Volatility Forecasting	Aug 2025
Model Developer, XJTLU	Suzhou, China
– Implemented temporal and positional encodings for multi-horizon FX-volatility forecasting.	
– Proposed cross-attention with daily technical representations as queries and lower-frequency macroeconomic and geopolitical streams as keys and values.	
– Evaluated 1-, 5-, and 10-day forecasting horizons and prepared pooled and horizon-specific comparisons.	
– Produced predicted-versus-actual plots, model-pipeline figures, and Integrated Gradients analysis of the 2024 U.S. election period.	
Sequential Suicide-Risk Prediction from Social Media	Nov 2025
Lead Student Researcher, First Author of ICSC Poster, XJTLU	Suzhou, China
– Built the RSD-15K pipeline from data acquisition and exploratory analysis through preprocessing, model training, and user-level evaluation.	
– Developed a hierarchical architecture combining Twitter-RoBERTa, Bi-GRU, and attention.	
– Applied emoji and token normalization, head-tail retention, and weighted focal loss for imbalanced risk categories.	
– Achieved weighted F1 0.46 versus 0.43 for a RoBERTa baseline under five-fold user-level evaluation.	

Mathematical Modeling and Selected Projects

The Fairness Paradox: A Stratified Hybrid-Merit Model

Jan 2026

Team Member, Mathematical Contest in Modeling

COMAP

- Contributed to a team modelling submission examining fairness in competitions that combine judges' scores with audience votes.
- Contributed to a team modelling submission combining Softmax-based inverse inference, counterfactual comparison of rank- and percentage-based scoring, and Kruskal–Wallis analysis.
- Helped formulate the team's proposed Stratified Hybrid-Merit Model. Named member of a 2026 MCM Finalist team.

Grouped NIPT Timing Modelling and Fetal-Abnormality Classification

Sep 2025

Statistical Modeling Contributor, CUMCM 2025

Jiangsu, China

- Studied grouped NIPT testing-time selection and fetal-abnormality classification for CUMCM 2025 Problem C.
- Contributed GLM/GAM analysis of BMI and gestational-age covariates and a BMI-grouped testing-time recommendation framework.
- Contributed logistic classification with ROC/Youden thresholding for T21, T18, and T13. The team received Second Prize in the Jiangsu Division, Undergraduate Group.

Au₂₀ Scientific Visualization Support

Oct 2025

Visualization and Reporting Contributor, XJTLU

Suzhou, China

- Built Python utilities for Au₂₀ atomic-cluster structures and bond diagrams. [Code](#).
- Rendered animation frames in Python and assembled them into LaTeX/PDF presentation sequences.

Home Repair Assistant Robotics

Jun 2025 – Aug 2025

Engineering Contributor, XJTLU SURF

Suzhou, China

- Configured the ROS 2 runtime and verified node-level execution across project modules.
- Deployed and validated the codebase on NVIDIA Jetson Orin Nano for the project demonstration.

Selected Honors and Awards

Finalist	2026
Mathematical Contest in Modeling	The Fairness Paradox: A Stratified Hybrid-Merit Model
Jiangsu Provincial Second Prize	2025
Contemporary Undergraduate Mathematical Contest in Modeling	Grouped NIPT Timing Modelling
Grand Prize, Top 1 out of 50+	2025
XJTLU Research-led Learning Symposium	Au ₂₀ Delta-Learning
First Prize	2025
XJTLU Research-led Learning Symposium	Grouped NIPT Timing Modelling
Second Prize	2025
XJTLU Research-led Learning Symposium	Mixed-Frequency FX Forecasting
Second Prize	2025
XJTLU Research-led Learning Symposium	Sequential Suicide-Risk Modelling
Jiangsu Provincial Third Prize	2025
“Keyou Cup” University Intelligent Robot Creativity Competition	Home Repair Assistant
Excellent Poster Award	2025
Summer Undergraduate Research Fellowship	XJTLU

Selected Presentations

International Conference on Social Computing	2025
Poster Presenter	ICSC 2025
– Presented <i>Public Mental Health Monitoring: Detecting Suicide Risk in Sequential Social Media</i> .	
– Communicated the sequential modelling approach, user-level evaluation protocol, and poster findings.	
XJTLU Research-led Learning Symposium	Nov 2025
Poster and Oral Presenter	XJTLU
– Advanced through poster screening and delivered a 10-minute oral presentation followed by a 5-minute Q&A.	
– Presented hierarchical suicide-risk modelling and received Second Prize for the project.	

Technical Skills

Programming and tools: Python, C/C++, Git, and LaTeX.

Python ecosystem: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, Pandas, Matplotlib, and Pygame.

Machine learning: Transformers, attention mechanisms, mixed-frequency and multi-horizon forecasting, cross-validation, and imbalanced classification.

NLP and data methods: RoBERTa encoding, Bi-GRU sequence modelling, statistical modelling, ROC analysis, and scientific visualization.

Languages: Chinese, native. English, English-medium degree programme with an official EMI letter available.